



SPEED CHANGE LANES LENGTHS WITH COMPLEMENTARY SPEED VALUES



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Lengths of Entrance Ramp Speed Change Lanes (m)

Highway Design Speed (km/h)	Entrance Curve Design Speed (km/h)											
	Stop	20	30	40	50	60	70	80	90	100	110	120
	Initial Speed (km/h)											
	0	20	28	35	42	51	63	70	77	85	91	98
Minimum Acceleration Length (m)												
50	60	50	30	-	-	-	-	-	-	-	-	-
60	90	60	55	30	-	-	-	-	-	-	-	-
70	155	135	130	110	80	-	-	-	-	-	-	-
80	225	205	200	180	150	90	-	-	-	-	-	-
90	305	290	285	265	230	170	100	-	-	-	-	-
100	385	370	360	335	300	250	175	90	-	-	-	-
110	465	440	430	405	375	320	245	150	48	-	-	-
120	545	530	515	490	460	410	325	245	156	46	-	-
130	625	605	655	571	540	491	415	340	218	109	44	-
140	703	687	793	652	624	572	507	438	350	245	155	37
150	819	806	945	776	750	700	646	581	492	392	305	190
160	987	977	965	940	928	877	787	726	657	570	496	397

Notes:

- *Taper type acceleration lanes should be used for high speed (80 km/h or greater) and high volume entrances. Otherwise; Parallel type could be used.
- *Generally if design speed exceeds 110 km/h or acceleration length exceeds 400 m, taper rate of 70:1 shall be used.
- *For design speeds greater than or equal to 80 km/h, acceleration taper rate is 50:1 and 25:1 taper could be used for lower speeds.

References:

MOT Highway Design Manual - Volume II - Book 1 of 2
 TxDOT 10/2014 Roadway Design Manual.

Lengths of Exit Ramp Speed Change Lanes (m)

Highway Design Speed (km/h)	Exit Curve Design Speed (km/h)											
	Stop	20 (25)	30	40	50	60	70	80	90	100	110	120
	Minimum Deceleration Length (m)											
50	76	(60)	56	46	-	-	-	-	-	-	-	-
60	96	(88)	76	66	48	-	-	-	-	-	-	-
70	114	(102)	98	86	70	54	-	-	-	-	-	-
80	132	(122)	118	106	92	80	60	-	-	-	-	-
90	152	(142)	138	128	118	106	84	60	-	-	-	-
100	168	(160)	156	148	138	126	104	82	56	-	-	-
110	182	(176)	172	164	154	140	120	100	78	52	-	-
120	200	195	185	175	170	155	140	120	102	78	58	-
130	225	220	210	201	195	180	160	141	116	92	73	60
140	248	241	234	226	217	202	178	162	144	121	103	80
150	271	264	258	250	241	227	204	189	172	150	132	110
160	309	303	297	290	282	268	248	233	216	196	180	159

Notes:

*Taper type deceleration lanes should be used for high speed (80 km/h or greater) and/or high volume exits. Otherwise, parallel type could be used.

*Where providing desirable deceleration length is impractical, it is acceptable to allow for a moderate amount of deceleration (10 km/h) within the through lanes and to consider part of the taper as part of the deceleration length.

*Deceleration taper rate is 25:1

References:

*MOT Highway Design Manual - Volume II - Book 1 of 2

*TxDOT 10/2014 Roadway Design Manual.

Speed Change Lane Adjustment Factors as a Function of a Grade

Deceleration Lanes

(Ratio of Length on Grade to Length on Level*)

Design Speed of Highway (km/h)	3 to 4 % Upgrade	3 to 4 % Downgrade	5 to 6% Upgrade	5 to 6% Downgrade	
All speeds	0.9	1.2	0.8	1.35	

Acceleration Lanes

(Ratio of Length on Grade to Length on Level*)

Design Speed of Turning Roadway Curve (km/h)

Design Speed of Highway (km/h)	30	40	50	60	70	80	All Speeds
Longitudinal Slope	3 to 4 % Upgrade						3 to 4% Downgrade
50	1.2	----	----	----	----	----	0.7
60	1.24	1.26	----	----	----	----	0.65
70	1.29	1.31	1.34	----	----	----	0.6
80	1.33	1.36	1.38	1.41	----	----	0.6
90	1.38	1.41	1.44	1.47	1.51	----	0.6
100	1.43	1.47	1.5	1.54	1.58	1.62	0.6
110	1.48	1.5	1.59	1.64	1.7	1.75	0.6
120	1.5	1.5	1.6	1.7	1.7	1.8	0.6
130	1.5	1.5	1.6	1.7	1.7	1.8	0.6
140	1.57	1.57	1.67	1.81	1.79	1.9	0.55
150	1.6	1.6	1.7	1.86	1.83	1.94	0.53
160	1.63	1.63	1.73	1.9	1.86	1.99	0.52
Longitudinal Slope	5 to 6% Upgrade						5 to 6% Downgrade
50	1.24	----	----	----	----	----	0.6
60	1.31	1.37	----	----	----	----	0.55
70	1.4	1.49	1.58	----	----	----	0.55
80	1.48	1.61	1.73	1.85	----	----	0.55
90	1.6	1.76	1.91	2.05	2.2	----	0.55
100	1.72	1.9	2.08	2.25	2.44	2.62	0.5
110	1.85	2.05	2.25	2.45	2.64	2.85	0.5
120	2.3	2.3	2.5	3	3.2	3.5	0.5
130	2.6	2.6	2.8	3.4	3.6	4	0.5
140	2.55	2.55	2.82	3.53	3.92	4.42	0.44
150	2.7	2.7	3	3.81	4.29	4.48	0.42
160	2.86	2.86	3.18	4.09	4.65	5.34	0.41

* Ratio from this table multiplied by length of level acceleration/deceleration distances gives length of speed-change lane on grade